Lecture Notes

on Interpolation, Extrapolation, and Correlation

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1 Interpolation and Extrapolation

1.1 Definition (Interpolation):

Interpolation is the process of estimating or predicting the value of a dependent variable for an independent variable that lies within the range of the observed data points.

i.e, $x \in [2, 12]$ (applicable only for the example below)

1.2 Definition (Extrapolation):

Extrapolation is the process of estimating or predicting the value of a dependent variable for an independent variable that lies outside the range of the observed data points.

i.e, $x \in (-\infty, 2) \cup (12, \infty)$ (applicable only for the example below)

Example: Recall the example from the last lecture, a farmer wants to examine whether the amount of fertilizer used (x, in kg) affects the yield of crops (y, in quintals). The data collected from 6 plots of land is:

x (Fertilizer, kg)	y (Observed Yield)
2	40
4	55
6	65
8	70
10	85
12	95

Where, we saw together in class while working in Google Sheets, the regression coefficients (slope and intercept for the least squares fitted line) are:

$$m \approx 5.29, \qquad b \approx 31.33$$

so the fitted regression line is

$$y = 5.29x + 31.33.$$

Questions:

1. Use the regression line to **interpolate** the yield when the farmer uses $x = 7$ kg of fertilizer.

Q: for $x = 7$, $y = ?$

$$y(7) = y\big|_{x=7} = (5.29 \times 7) + 31.33$$

$$= 68.36$$

Thus the required yield is 68.36 quintal.

2. Use the regression line to **extrapolate** the yield when the farmer uses $x = 15$ kg of fertilizer.

$$y = 5.29x + 31.33$$

$$y = (5.29 \times 15) + 31.33$$

$$y = 79.35 + 31.33$$

$$y = 110.68 \text{ quintal}$$

3. If the farmer obtains a crop yield of $y = 90$ quintals, use the regression line to estimate the amount of fertilizer used (x).

$x = ?$ for $y = 90$.

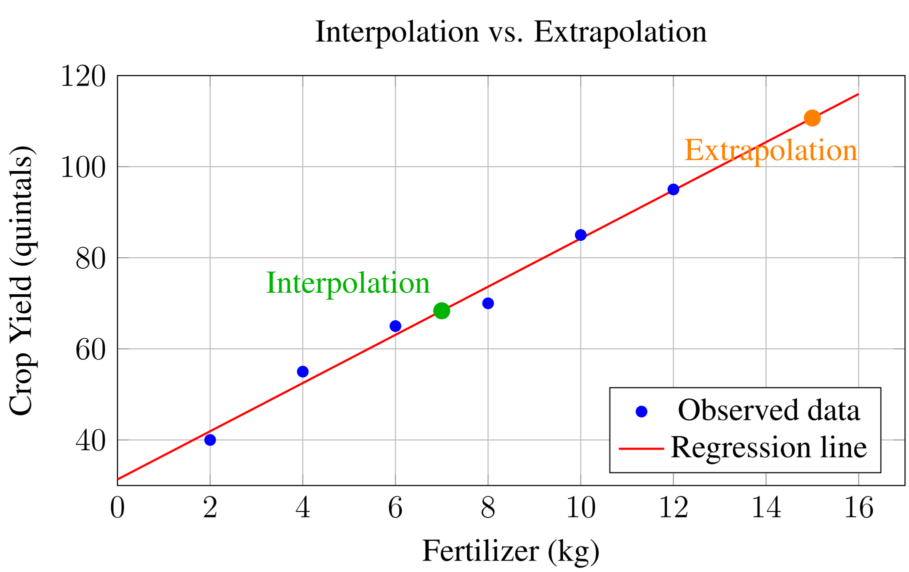
From, $y = 5.29\,x + 31.33$

$$\Rightarrow 5.29x = y - 31.33$$

$$\Rightarrow x = \frac{y - 31.33}{5.29}$$

$$\therefore x\Big|_{y=90} = \frac{90 - 31.33}{5.29} = 11.09 \text{ kg}.$$

Note: In general, we might trust an **interpolation** more than an **extrapolation**, because interpolation predicts values within the observed data range (where we have evidence of the trend), while extrapolation predicts values outside the observed range (where the trend may no longer hold).



2 Correlation and Correlation Coefficient

ρ (rho)

2.1 Definition (Correlation):

Correlation is a statistical measure that describes the strength and direction of the linear relationship between two variables. It tells us whether an increase in one variable tends to be associated with an increase (positive correlation), a decrease (negative correlation), or no consistent change (no correlation) in the other variable.

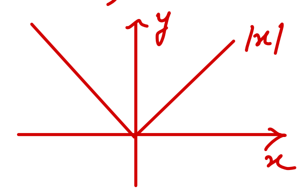
2.2 Definition (Correlation Coefficient):

The correlation coefficient, denoted by ρ, is a numerical value that quantifies the degree of correlation between two variables.

Properties:

- $-1 \leq \rho \leq 1$.
- $\rho = 1$: perfect positive correlation.
- $\rho = -1$: perfect negative correlation.
- $\rho = 0$: no linear correlation.
- Anything less than 0.5 in absolute value is usually considered too weak to suggest a meaningful correlation. $|\rho| < 0.5$

$|x|, x \in \mathbb{R}$

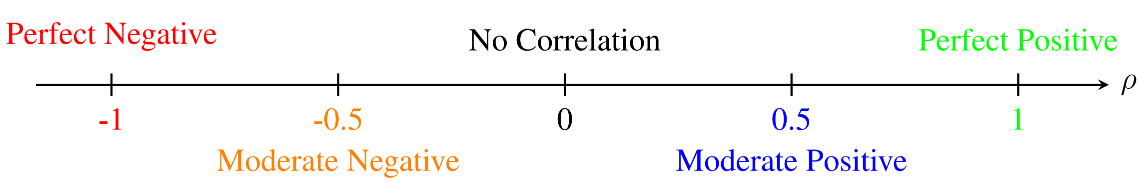


$|-2| = 2$

$|0| = 0$

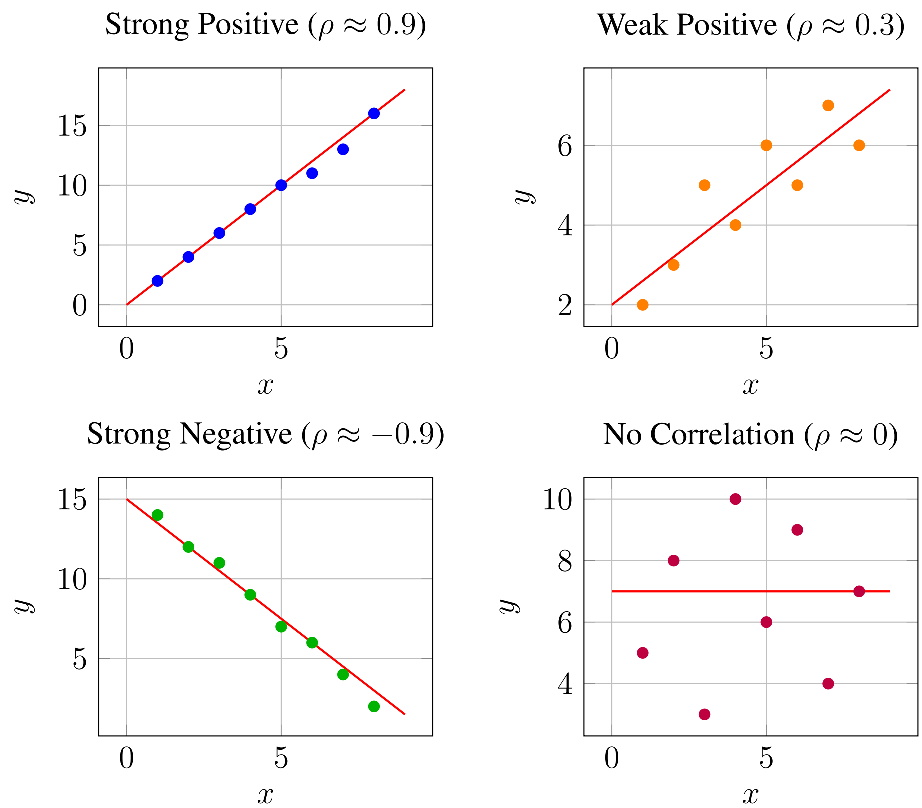
Rule of Thumb for Strength of Correlation:

- Weak: $|\rho| < 0.5$
- Moderate: $0.5 \leq |\rho| < 0.7$
- Strong: $|\rho| \geq 0.7$



Note: You are not required to perform any numerical calculations of R^2 (total residual squared) or ρ (correlation coefficient) using formulas. However, you must understand the underlying concepts and their significance, as discussed so far.

Graphical Representation of Correlation Coefficients



Relation Between Correlation Coefficient and Slope

The slope m of the regression line and the correlation coefficient ρ are connected only in terms of **sign**:

- If $\rho > 0$, then the slope $m > 0$ (the line rises from left to right).
- If $\rho < 0$, then the slope $m < 0$ (the line falls from left to right).
- If $\rho = 0$, then the slope $m \approx 0$ (no linear relationship).

Important: The value (magnitude) of the slope has nothing to do with the value of the correlation coefficient.

Correlation measures strength of linear association, while slope measures rate of change.

Correlation vs. Causation

Correlation measures the strength and direction of a linear relationship between two variables. **Causation**, on the other hand, means that changes in one variable directly cause changes in the other.

Correlation does not imply causation.

☛ **Examples:**

- Ice cream sales and drowning incidents may be positively correlated, but the cause is the hot summer weather, not ice cream.
- Shoe size and reading ability in children may be correlated, but the underlying cause is age.

Hence, while correlation and regression can suggest patterns, they should never be interpreted as proof of cause-and-effect without further evidence.

3 Outliers

3.1 Definition (outlier):

An **outlier** is a data point that lies far from the overall pattern of the data. Outliers may arise due to unusual conditions, measurement errors, or genuine rare events, and they can significantly affect correlation and regression analysis.

☛ **Example (Car Age vs. Resale Value):** A car dealer records the age of cars (x, in years) and their resale value (y, in \$1000).

x (Car Age, years)	y (Resale Value, \$1000)
1	25
2	22
3	20
4	18
5	15
6	13
7	11
8	50 (Outlier)

Here, the 8-year-old car is unusually expensive compared to the general trend, perhaps it's a rare vintage model, making it an outlier.

Graphical Representation: